to Einstein’s newest heresy.

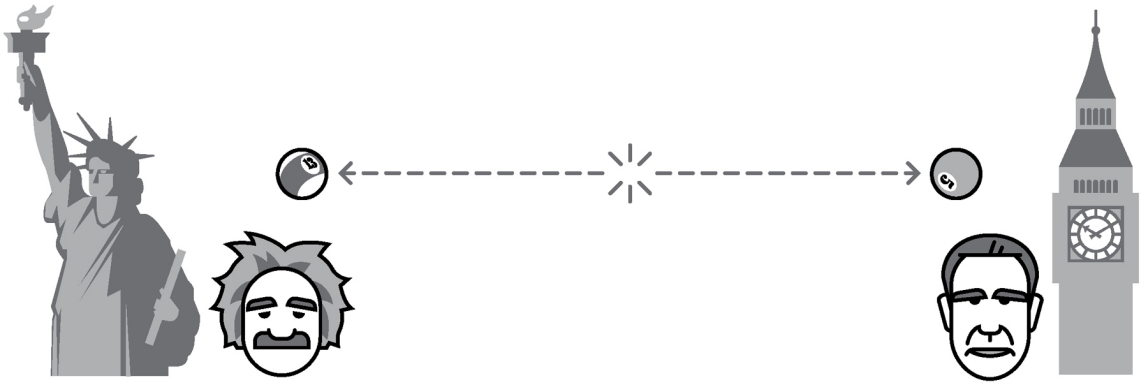


Figure 3.4. The EPR experiment. Two billiard balls collide and fly off in opposite directions. When Albert measures the momentum of his billiard ball, he instantly infers the momentum of Niels’s billiard ball, even though he is in New York and Niels is in London. Either Niels’s billiard ball already had that momentum in London before Albert made his measurement in New York, or there is “spooky action at a distance,” instantly connecting the two balls across the Atlantic.

“This onslaught came down upon us as a bolt from the blue. Its effect on Bohr was remarkable,” said Léon Rosenfeld. “As soon as Bohr had heard my report of Einstein’s argument, everything else was abandoned; we had to clear up such a misunderstanding at once.” Bohr immediately began crafting a reply with Rosenfeld’s help. Normally a painfully slow writer, Bohr managed to crank out a paper replying to EPR in six weeks—“an astonishing speed” for Bohr, according to Rosenfeld—and sent it off to *Physical Review,* the same journal where the EPR paper had appeared.

Bohr considered the EPR thought experiment carefully in his reply. He agreed that measuring A’s momentum could not “mechanically” disturb B—there was “no question” of that. But, he maintained, there was still “the question of *an influence on the very conditions which define the possible types of predictions regarding the future behavior of the system.*” Unfortunately, it’s not clear what distinction Bohr was trying to draw between “mechanical disturbance[s]” on one hand and “influences” on the other. Was he saying that measuring A could immediately affect B? Maybe. Did he think that quantum physics therefore had to be nonlocal? Maybe